

NAKHONPATHOM, Thailand

WMO: 484510

Lat: 14.0117N

Lon: 99.9700E

Elev: 32

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.3	64.9	53.8	61.7	68.2	57.6	70.8	71.4	10.1	78.2	9.4	78.5	1.2	N/A	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.4	99.2	80.0	97.3	79.9	95.9	79.9	82.8	91.9	82.2	91.4	81.7	90.8	3.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.8	85.8	80.1	156.6	85.2	79.7	154.3	85.0	46.8	92.7	46.1	91.7	45.5	91.1	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.0	7.4	DB	57.8	101.7	3.9	2.3	55.0	103.4	52.7	104.7	50.5	106.0	47.7	107.7	(4)
			WB	55.8	84.7	4.4	1.3	52.6	85.6	50.0	86.3	47.5	87.1	44.3	88.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	77.6	81.9	84.9	87.4	87.0	85.5	84.5	84.4	84.0	82.9	81.0	77.6	(6)
(7)		DBStd	4.26	4.35	3.39	3.60	3.07	2.75	1.84	1.90	1.67	1.82	1.95	2.79	4.18	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12130	856	892	1082	1123	1148	1064	1070	1068	1020	1021	930	856	(10)
(11)		CDD65	6655	391	472	617	673	683	614	605	603	570	556	480	391	(11)
(12)		CDH74	75052	3527	5145	7595	8864	8647	7344	6947	6808	6243	5925	4673	3334	(12)
(13)		CDH80	33349	1323	2355	3786	4836	4404	3287	2903	2799	2516	2253	1719	1168	(13)
(14)	Wind	WSAvg	2.0	1.7	1.9	2.5	2.4	2.2	2.0	2.1	2.2	1.9	1.6	1.9	2.0	(14)
(15)	Precipitation	PrecAvg	42.2	0.3	0.6	1.4	2.6	5.4	4.0	4.3	4.4	9.1	8.2	1.7	0.2	(15)
(16)		PrecMax	55.0	2.1	3.6	6.9	9.2	10.9	7.6	7.2	7.2	17.1	14.4	5.2	1.6	(16)
(17)		PrecMin	29.3	0.0	0.0	0.0	0.0	0.7	0.5	2.0	2.0	4.1	1.1	0.0	0.0	(17)
(18)		PrecStd	6.0	0.4	0.8	1.6	2.1	2.3	1.7	1.6	1.3	3.3	3.2	1.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	95.7	99.1	101.9	101.7	97.7	96.2	95.4	95.2	94.2	93.1	93.0	(19)
(20)		0.4%	MCWB	77.8	78.2	76.3	79.5	80.9	81.0	79.8	80.1	80.7	80.2	80.2	78.0	(20)
(21)		2%	DB	91.0	94.2	96.9	99.7	99.0	95.4	94.0	93.6	93.6	92.2	91.4	91.0	(21)
(22)		2%	MCWB	76.9	77.8	77.6	80.1	81.1	80.9	79.9	80.0	80.5	80.1	78.9	77.6	(22)
(23)		5%	DB	89.4	92.7	95.5	97.8	96.8	93.5	92.3	92.1	92.0	90.7	90.0	89.0	(23)
(24)		5%	MCWB	76.2	77.4	78.4	80.1	81.0	80.6	79.6	79.8	80.2	79.8	78.5	76.9	(24)
(25)		10%	DB	87.3	91.0	93.9	95.9	94.8	91.8	90.6	90.5	90.4	89.3	88.2	86.4	(25)
(26)		10%	MCWB	75.6	77.1	78.5	80.3	80.8	80.4	79.3	79.6	79.9	79.6	77.8	75.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	81.1	81.9	83.6	84.2	82.8	82.0	82.2	82.5	82.2	81.8	80.1	(27)
(28)		0.4%	MCDWB	88.9	91.7	92.3	92.6	93.1	92.2	91.0	90.3	91.0	89.1	88.9	88.0	(28)
(29)		2%	WB	79.0	80.1	81.0	82.6	82.8	81.9	81.0	81.3	81.6	81.5	80.9	78.9	(29)
(30)		2%	MCDWB	87.5	90.0	90.8	92.2	92.2	91.4	90.0	89.6	89.9	88.4	88.3	87.4	(30)
(31)		5%	WB	78.2	79.4	80.5	81.9	82.2	81.5	80.5	80.7	80.9	80.9	80.1	77.9	(31)
(32)		5%	MCDWB	85.9	88.5	89.8	91.4	91.7	90.5	89.3	89.0	89.1	87.9	87.2	85.9	(32)
(33)		10%	WB	77.2	78.6	79.9	81.3	81.5	80.9	79.9	80.0	80.4	80.3	79.2	76.9	(33)
(34)		10%	MCDWB	84.0	87.0	88.5	90.6	91.0	89.5	88.0	87.9	88.2	87.0	85.3	84.3	(34)
(35)	Mean Daily Temperature Range		MDBR	18.1	18.6	17.5	17.4	15.8	14.0	13.2	13.5	13.6	12.9	14.0	16.3	(35)
(36)		5% DB	MCDBR	20.3	19.8	19.6	19.3	18.5	16.2	15.5	15.5	15.8	15.1	16.3	18.8	(36)
(37)		5% DB	MCWBR	7.1	5.6	5.0	4.8	4.7	4.8	4.6	4.7	5.1	4.8	5.2	6.5	(37)
(38)		5% WB	MCDBR	17.2	17.7	17.7	17.8	16.7	14.8	14.3	14.2	14.7	13.9	14.5	17.0	(38)
(39)		5% WB	MCWBR	6.2	5.4	5.2	5.1	5.2	4.9	4.7	4.8	5.2	4.9	5.3	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.540	0.604	0.642	0.570	0.489	0.450	0.443	0.444	0.469	0.579	0.527	0.517	(40)	
(41)		taud	1.949	1.823	1.764	1.980	2.265	2.373	2.394	2.378	2.297	1.946	2.061	2.031	(41)	
(42)		Ebn at Noon	240	231	226	243	259	266	269	272	266	235	243	243	(42)	
(43)		Edh at Noon	58	68	74	59	44	39	38	39	43	59	51	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1546	1695	1763	1910	1730	1609	1514	1514	1529	1471	1514	1501	(44)	
(45)		RadStd	92	73	134	114	140	111	101	107	99	92	103	71	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	+0.45	+0.49	N/A	N/A	+155	+155	(47)

Nomenclature: See separate page